The University of Zambia

in association with

ZCAS University

BFS322 TREASURY MANAGEMENT

FINAL EXAMINATION

30 November 2020

12:30 – 15:30 HOURS

TIME ALLOWED: **WRITING – THREE HOURS**

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer Three (3) questions from this section.
3. This question paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

The Ministry of Finance

The Ministry of Finance derives its mandate from the Constitution of Zambia (Amendment) Act, 2016 Part XVL. It is also guided by statutory and portfolio functions stipulated in the Government Gazette Notice No. 76 of 2016. The Ministry of Finance is charged with the responsibility of economic management to foster sustainable national development".

The Ministry is headed by a Minister whose mandate is drawn from the Minister of Finance (Incorporation) Act, Chapter 349 of the Laws of Zambia. The administrative and technical team is headed by the Secretary to the Treasury who is assisted by four Permanent Secretaries.

Mission Statement

"To efficiently and effectively coordinate national planning and economic management, mobilise and manage public resources in a transparent and accountable manner for sustainable national development".

Economic Management Department

The role of the Economic Management Department (EMD) is to put in place or design a macroeconomic policy framework that would facilitate sustainable economic growth and stability. In this regard, EMD is responsible for the development of effective monetary and fiscal policies that would guide domestic resource mobilization, as well as Government medium and long-term expenditure. Furthermore, the Department is also mandated to spearhead economic monitoring and analysis in consultation with national financial institutions and other government agencies. It is also responsible for undertaking and coordinating external financial resource mobilization.

(source: www.mof.zm)

(i) Critically analyse the role that the Ministry of Finance plays in managing the country's Treasury. **(20 marks)**

(ii) Discuss how the Ministry coordinates with the Bank of Zambia in its management of the country's treasury. **(10 marks)**

(iii) The Economics Department at the Bank of Zambia works closely with the Ministry of Finance in managing the country's treasury. Identify the divisions under the Economics Department at the Bank of Zambia and describe the role these divisions play in treasury.

(10 marks)

(Total: 40 marks)

SECTION B: **Attempt any THREE questions in this section**

Question 2

Luano Resources Limited is a small-scale contractor that supplies to the mining houses on the Copperbelt in Zambia. To finance its working capital needs, the company enjoys an overdraft facility of K100,000 which is now due for review. The average balances on the current account for the last 12 months are shown below:

Month	**Balance on current account**	**Excess over limit**
January	K102,000	0
February	K50,400	0
March	K1,020	0
April	(K55,200)	0
May	(102,000)	K2,000
June	(105,000)	K5,000
July	(K101,000)	K1,000
August	(K105,000)	K5,000
September	(K104,000)	K4,000
October	(K106,000)	K6,000
November	K105,000)	K5,000
December	K105,000)	K5,000

(i) Discuss the advantages and disadvantage to Luano Resources Limited of using an overdraft to finance its working capital. **(8 marks)**

(ii) The bank has refused to renew the overdraft for a further 12 months. From the figures shown above, what would you say could be the reason for this? **(2 marks)**

(iii) The bank has offered to discount selected invoices that Luano Resources has outstanding with the mining companies. Write brief notes on this form of financing and contrast it with factoring financing. **(10 marks)**

Question 3

ABC Ltd. uses EOQ logic to determine the order quantity for its various components and is planning its orders. The Annual consumption is 80,000 units, Cost to place one order is ZMK 1,200, Storage cost per unit is ZMK 50.

(i) Calculate: EOQ, number of orders per year, ordering cost, storage cost and total cost of inventory. **(10 marks)**

(ii) ABC is considering switching to a JIT system of managing its inventory. Critically analyse this system and identify the merits and demerits of implementing the JIT system. **(10 marks)**

Question 4

ABC Company can borrow from its bank at a fixed rate of 4% or at a variable rate of LIBOR+1. XYZ on the other hand can borrow from its bank at a variable rate of LIBOR+2 or at a fixed rate of 6%. ABC Company has got a comparative advantage borrowing fixed but would like to borrow at a floating rate. XYZ would like to borrow at the floating rate.

ABC Company and XYZ Company enter into one-year interest rate swap with a nominal value of $1 million. ABC offers XYZ a fixed annual rate of 5% in exchange for a rate of LIBOR plus 1%, since both parties believe that LIBOR will be roughly 4%.

(i) Give reasons why the two companies would want to enter into this agreement? **(5 marks)**

(ii) Identify the risks to both parties of entering into this arrangement and explain how they can mitigate the risk. **(5 marks)**

(iii) With the aid of a diagram, illustrate the amount of interest that will be exchanged between the two parties under the agreement. **(10 mark)**

Question 5

Average Exchange rates: 10-11-2020		
Currency	**Buying**	**Selling**
USD	20.8251	20.8751
GBP	27.5703	27.6428
EUR	24.5882	24.6514
ZAR	1.3386	1.3426

(Source: Bank of Zambia)

(i) Given that the yield on the 91 day treasury bill rate in Zambia is 14.000% and the yield on the 3 month treasury rate in the U.S.A is 1.66%, calculate the 90 day forward rate for USD/ZMK using the rate table above (Note: use the USD/ZMK mid-rate for spot). **(6 marks)**

(ii) Discuss the three types of foreign exchange exposure that a firm dealing in a foreign currency might be exposed to. **(6 marks)**

(iv) Suggest any two (2) business practices that a firm can use to manage its foreign currency risk that do not involve the use of financial derivative instruments. Discuss the limitations of the practices that you choose. **(8 marks)**

END OF TEST

MARKING KEY

Question 1

(i) Role that MOF plays in managing the Country's treasury:

- Formulates and implements fiscal and monetary policy
- In charge of budget preparation and execution
- Manages the country's financial resources
- In charge of disbursing funds to the ministries for spending
- In charge of resource mobilisation e.g. through tax receipts
- In charge of public debt management

(ii) How the MOF coordinates with BOZ in Management of treasury:

- Coordinates in the implementation of monetary policy
- BOZ acts as fiscal agent of GRZ through issuance of TBs and Bonds
- BOZ maintains the data base for government securities
- MOF plays role in the development of the financial sector
- Government Ministries accounts are held at BOZ
- Government's main account (control account 99) is held at BOZ
- MOF coordinates with BOZ in the management of public debt
- BOZ acts as an advisor to government on financial and economic matters.

(iii) Divisions in the Economics department at BOZ and treasury role:

- Generates monetary policy information through four divisions:

a. Macroeconomic analysis- preparation of analytical and policy related reports on monetary, fiscal and other economic developments

b. Balance of Payments and External Debt- compilation and analysis of bop and external debt development

c. Information and Statistics- compilation of data on economic developments and dissemination of information to general public

d. Research- undertakes and co-ordinates research activities related to monetary and other economic issues

- Works closely with Economic Ministries and international organisations like IMF, World Bank, UNECA et

Question 2

(i) Advantages of Overdraft:

Flexibility

- Customer may draw funds only when required

Minimum documentation

- Legal documentation is fairly minimal when arranging an overdraft

Interest

- Paid on overdrawn amount rather than full facility
- If secured lower rate may be available because of lower risk to lender

Disadvantages of overdraft:

- Facility is repayable on demand
- Bank will demand collateral
- Borrower exposed to interest rate fluctuations
- Interest rate charged varies according to perceived risk

(ii) Reason for refusal to renew overdraft:

The account has persistently been operated above the limit of K100,000. This is an indication of a hardcore on the account, which shows that the customer was having difficulties servicing the overdraft.

(iii)

Invoice discounting	Factoring
Description: • Financier advances cash against outstanding invoices on reputable debtors	• Financier advances cash against outstanding invoices and in addition manages the customer sales ledger and credit
Contrast: • Financier only advances cash against invoices • Financier does not manage customers sales ledger nor are they involved in the administration of customers receivables • Facility is with recourse to the customer	• Involves the outsourcing of the credit management to the financing firm • Can be with or without recourse • The facility is non-confidential as the debtors will be made aware of the involvement of the factor

• Facility is confidential i.e. the company's debtors are not aware of the arrangement	

Question 3

(i) EOQ= $\sqrt{\frac{2ac}{h}}$

= $\sqrt{\frac{2 \times 80{,}000 \times 1{,}200}{50}}$

= 1,959 units

EOQ	Ordering Cost	Storage Cost	Total Cost
1,959	80,000/1,959 = 41 x 1200 =K49,005	1,959 x 50 =97,950	49,005+97950 = K146, 955

(ii) JIT system:

- JIT attempts to minimise inventory levels by reducing the costs or uncertainties that underlie the motives for holding inventory.
- JIT purchasing means that materials or components are received from the supplier just in time to use them

Objectives of JIT purchasing:

- Raw material stock reduction;
- Frequent deliveries with smaller orders from a smaller number of suppliers;
- Long term contracts with suppliers
- Quality assurance, with supplier becoming responsible for inspection of goods supplied
- Reduced inventory
- Savings in storage space required
- Increased customer satisfaction – elimination of waste leads to better quality product

Question 4

(i) Reasons why the two companies may want to enter into a swap arrangement:

- The parties may enjoy a comparative advantage borrowing in one type of interest rate as opposed to the other.
- The parties may want to manage their interest rate exposure by swapping their interest rate commitments.
- The parties may want to profit from the arrangement by entering into a speculative arrangement

(ii) Risks to the parties of entering into a swap agreement:

Interest rate risk

- The actual interest rate movements may not match the expectations of the parties
- For example, the counter party receiving a fixed rate payment stream loses if interest rates rise and vice versa

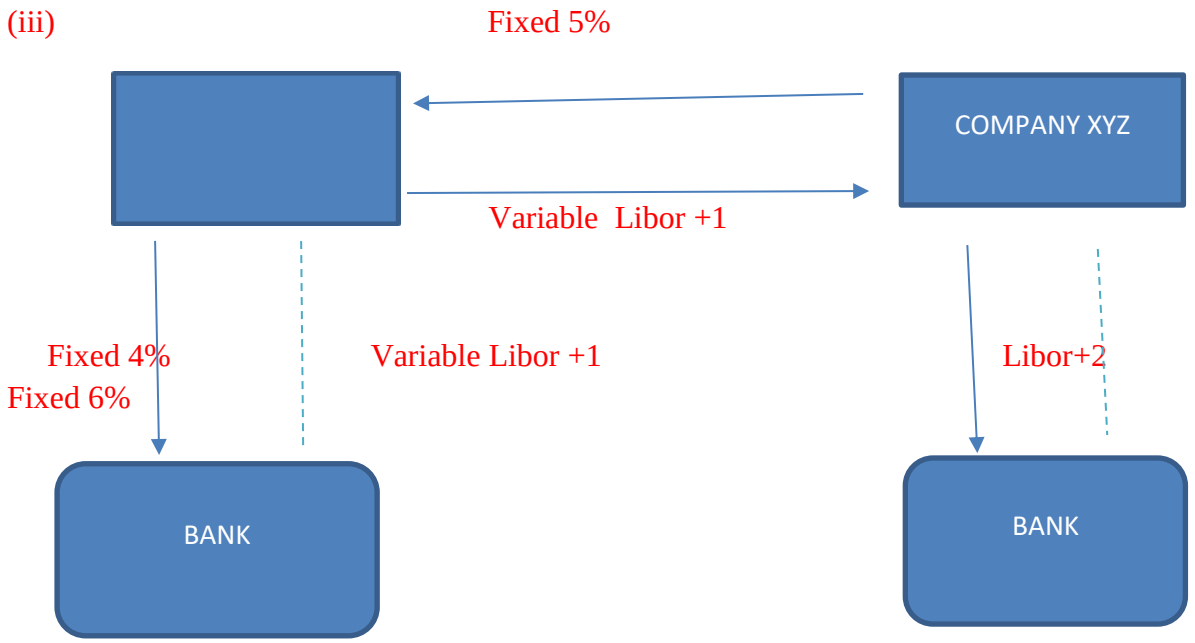
Counter party credit risk

- There is a risk that the counter party will default on their commitments.

Mitigation

- The parties can conduct the swap transaction through a bank intermediary to reduce the credit risk
- The parties may be required to pledge collateral as part of the transaction.
- The parties are required to sign swap agreements

(iii)



Company ABC

Pays fixed to bank = (4%)

Receives fixed from XYZ= 5%

Pays variable to XYZ = (5)%

Company XYZ

Pays variable to bank = (6%)

Receives variable from ABC = 5%

Pays fixed to ABC = (5%)

Interest per annum to be exchanged between the parties = 5% x $1,000,000 = $50,000

Question 5

(i) $TB_z = 14.000\%$

$TB_{us} = 1.66\%$

Mid rate USD/ZMK = 20.8251+20.8751/2 = 20.8501

Term = 90 days

$$F_{USD/ZMK} = \frac{(r^* - r) \times S}{(1+r)} + S$$

$$= 20.8501 \frac{(0.0166 - 0.14) \times 90/360}{1 + 0.14 \times 90/360} + 20.8501$$

$$= (20.8501 \times -0.122632) + 20.8501$$

$$= 18.2932$$

(ii) Types of FX exposure:

Economic risk

- Economic exposure directly impacts the value of a firm.
- That means, the value of the firm is influenced by the foreign exchange.
- The value of a firm is the function of operating cash flows and the assets it possesses.
- The economic exposure can have bearings on assets as well as operating cash flows. Identification and measuring of this exposure is a difficult task.
- Although, the asset exposure is still measurable and visible in books but the operating exposure has links to various factors such as competitiveness, entry barriers, etc which are quite subjective and interpretation of different experts may be different.

Transaction risk

- this exposure pertains to the exposure due to an actual transaction taking place in business involving foreign currency.
- If you have bought goods from a foreign country and payables are in foreign currency to be paid after 3 months, you may end up paying much higher on the due date as currency value may increase.
- This will increase your purchase price and therefore the overall costing of the product compelling the profit percentage to go down or even convert to lose.

- Transaction exposure normally occurs due to foreign currency debtors of sale, payment for imported goods or services, receipt / payment of dividend,

Translation risk

- This exposure is also well known as accounting exposure.
- It is because the exposure is due to the translation of books of accounts into the home currency.
- Translation activity is carried out on account of reporting the books to the shareholders or legal bodies.
- Gains or losses arising out of translation exposure do not have more meaning over and above the reporting requirements.
- Such exposure can even get reversed in the next year translation if currency market moves in the favorable direction.
- This kind of exposure does not require too much of management attention.

(iii) Business Practice that a firm may use to manage its FX exposure:

1. Insist on home currency payment
2. Currency surcharges
3. Get paid on time
4. Foreign currency loans
5. Sourcing Changes
6. Foreign Currency Accounts
7. Unilateral, Bilateral and Multilateral Netting Arrangements